

In-depth analysis of ISO7816 protocol-simple and easy to understand protocol detailed (1)-reset, character frame, and ATR

tags: Protocol analysis

1. Summary

The IC card must support the protocol of T=0 or T=1, but not both protocols at the same time, and the terminal must support the protocol of T=0 and T=1 at the same time.

- T=0 communication protocol is an asynchronous half-duplex character transmission protocol;
- T=1 communication protocol is an asynchronous half-duplex block transmission protocol;

In the ISO7816-3 standard, these two protocols are specified: T0 and T1.

The protocol used by the IC card is specified in TD1. If there is no TD1 in the reset response message, it means that the T=0 protocol is used for communication. After the reset response, the IC card and the terminal communicate with the protocol specified by the IC card.

2. Hardware interface

- 1) VCC: Power input (Type A: 5V, Type B: 3V)
- 2) GND: Ground
- 3) RST: Reset
- 4) I/O: input and output
- 5) CLK: Clock (Type A: 1~5MHz, Type B: 1~4MHz)

Note: The number of pins of different types of cards may be different.

3. Operation of reader and card

3.1 Introduction to ETU (Basic Time Unit)

The communication protocol defined by 7816-3 is basically a replica of RS232 and an improvement on this basis. In RS232, we have the concepts of 9600 baud rate, start bit, parity bit, and stop bit, which are completely retained in 7816-3, except that etu is introduced in 7816-3 and bps is not used. . The definition of etu can more accurately describe the responsibilities and role transitions of each data bit during the transmission process.

According to the definition, when the smart card is powered on and reset **$1 \text{ etu} = 372 / f$** , Where f = the clock frequency provided by the read-write device to the smart card through the CLK pin, usually between 1-5 MHz.

The unit of etu is the time unit second (milliseconds, microseconds), which is equivalent to the time required to transmit each data bit. The reciprocal of it is the data bits transmitted per second, which is bps. We take $f = 3.579545 \text{ MHz}$. Divide 3,575,545 by 372, and the result is equal to 9622.4 which is approximately 9600 bps.

General formula for etu calculation: **$1 \text{ etu} = (F / D) * (1 / f)$** , The values of F and D are set according to the agreement in the 7816-3 specification. For the value $F = 372$ at power-on reset, $D = 1$ is the default value F_d and D_d . If the smart card supports other rates, you need TA1 in the ATR to indicate the other values of F and D. For example, if

you set $F=372$, you can set D to 2/4 respectively, then the communication rate that the smart card can support can be respectively 19200/38400.

So why choose a clock frequency of 3.579545 MHz? Why is the default F value chosen as 372? First of all, 3.579545 MHz is the nominal value of commonly used quartz crystals, $9600 * 372 = 3.5712$ MHz, which is the closest. Besides, why is it 372, $372 = 12 * 31$. The standard 8051 single-chip microcomputer has 12 clock cycles per instruction cycle, and the timer/counter is also divided by 1/12 before counting. This setting makes it easier for the smart card chip to control the IO data communication. It can also be better matched with the PC serial port 9600 rate (of course this refers to the 1980s, for the current smart card chip design technology to achieve arbitrary frequency control is not a problem).

It can be seen that the selection of these parameters is to make better use of the existing standards and technologies at that time. It should be known that 7816-3 was formulated in 1989, and the serial port rate was set according to the mainstream PC configuration at the time. For 9600, use a 232 interface chip (such as Max232) plus a 3.579545MHz crystal oscillator, and then cooperate with a few 74 series gate circuits plus a 5 volt DC power supply and a few resistive components. This is the most basic RS232 serial port reading Card device.

PS:

- Fi Di: TA1 (high 4 bits, low 4 bits), see 4.3.1 for details.
- Fn Dn: PPS1 (high 4 bits, low 4 bits), see 4.3.1 for details.
- Fd Dd: Default value ($Fd = 372$, $Dd = 1$).

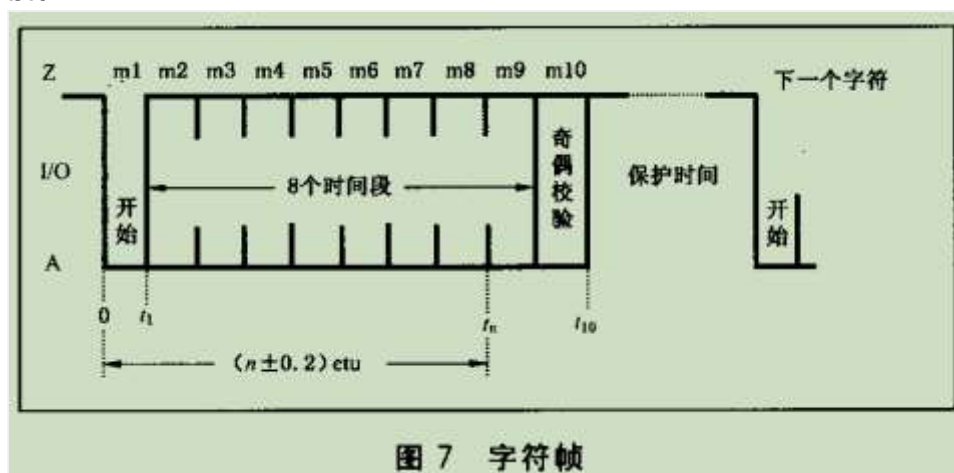
3.2 Character frame

Before the character, the I/O terminal should be in state Z. As shown in the figure below, a character should consist of 10 consecutive time periods. Each time period is either in state Z or in state A.

-The first time period m1 should be in state A; this time period is the "start bit".

— M2-m9 transports 1 byte in eight time periods.

— The last time period m10 should ensure the character parity bit and carry the "parity bit".



In each character, if the state changes at the end of the time period mn, the delay from the leading edge of the character to the trailing edge of mn should be $t_n = (n \pm 0.2) * \text{etu}$.

- The time starting point of the sender is the starting edge of the character. When searching for a character, the receiver periodically samples the I/O, and the sampling time should be less than 0.2etu. The receiver's starting point in time is between the last

observation point of the Z state and the first observation point of the A state.

- The receiver should confirm m1 before 0.7etu (receiver time), and then should receive m2 at $(1.5 \pm 0.2)\text{etu}$, and m3 at $(2.5 \pm 0.2)\text{etu}$, ..., at $(8.5 \pm 0.2)\text{etu}$ received m9, and received m10 at $(9.5 \pm 0.2)\text{etu}$. The character parity check is performed after the character frame is transferred.

Note: This allowable deviation ensures that the signal measurement area and the signal jump area are completely distinguished.

The delay between the starting edges of two consecutive characters should be at least 12 etu, that is, the duration of a character $(10 \pm 0.2)\text{etu}$ plus the guard time. During the guard time, both the interface device and the card should maintain the receiving state, so the I/O state is Z.

During the response to reset, the delay between the start edges of two consecutive characters sent by the card should not exceed 9600 etu. This maximum value is called the "initial waiting time".

4 Communication process

① Activate the circuit.

② The cold reset starts the card response, and the card communicates with the reader.

③ Release the circuit.

As the contacts of the IC card are actually turned on, the terminal must initiate a cold reset.

If the returned byte of the cold reset response of the IC card does not meet the requirements in the previous section, or the reset response of the IC card is not completed within the initial etu of 19,200, the terminal does not need to cancel the card use process immediately, but sends a Warm reset signal.

If the response meets the requirements and it is sent back within the initial etu of 19,200, the terminal must use the returned parameters for the card operation process.

If the terminal initiates one of the above-mentioned warm resets, and the bytes returned by the IC card's response to the warm reset still do not meet the requirements in the above section, or the IC card's reset response cannot be completed within the initial etu of 19,200, Then the terminal will execute the contact sequence of releasing the IC card, thereby canceling the card use process.

If the response to the hot reset meets these regulations and is sent back within 19,200, the terminal must use the returned parameters to process the card operation process. Regardless of whether it is a cold reset or a warm reset, if the time between the rising edges of the start bit of the adjacent two bytes returned by the IC card in response exceeds the 9600 initial etu, the terminal must execute the release contact sequence, thus Cancel the card use process.

4.1 Activation process

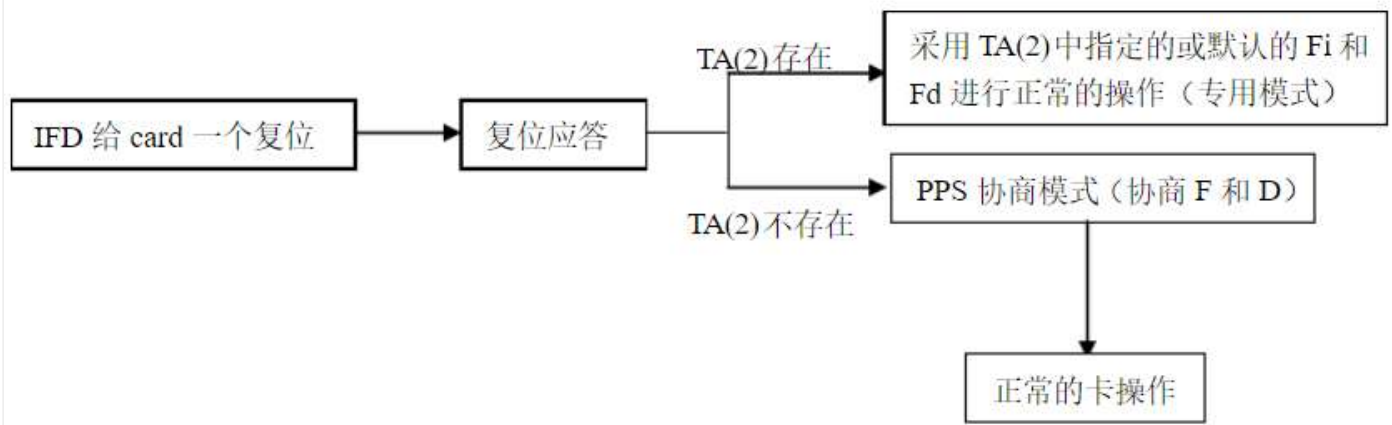
① RST sets the state L.

② VCC is powered on with 5V (Type A) or 3V (Type B).

③ I/O is set to receive state.

④CLK provides 1₅MHz (Class A) or 1₄MHz (Class B).

操作流程:



4.2 Cold reset

1. VCC power supply, RST state is at L, IO state is at L.
2. When the stable CLK starts, IO will change state to H within 200 clock cycles.
3. When a stable CLK is provided, RST will remain in the L state for at least 400 clock cycles.
4. After RST enters state H, the reset response starts within 400 to 40,000 clock cycles from the rising edge of RST's H state.
5. If after the rising edge of RST state H, more than 40,000 clock cycles, the card's reset response does not start, the reset fails, the signal on RST will return to state L, and the device will follow the release sequence.

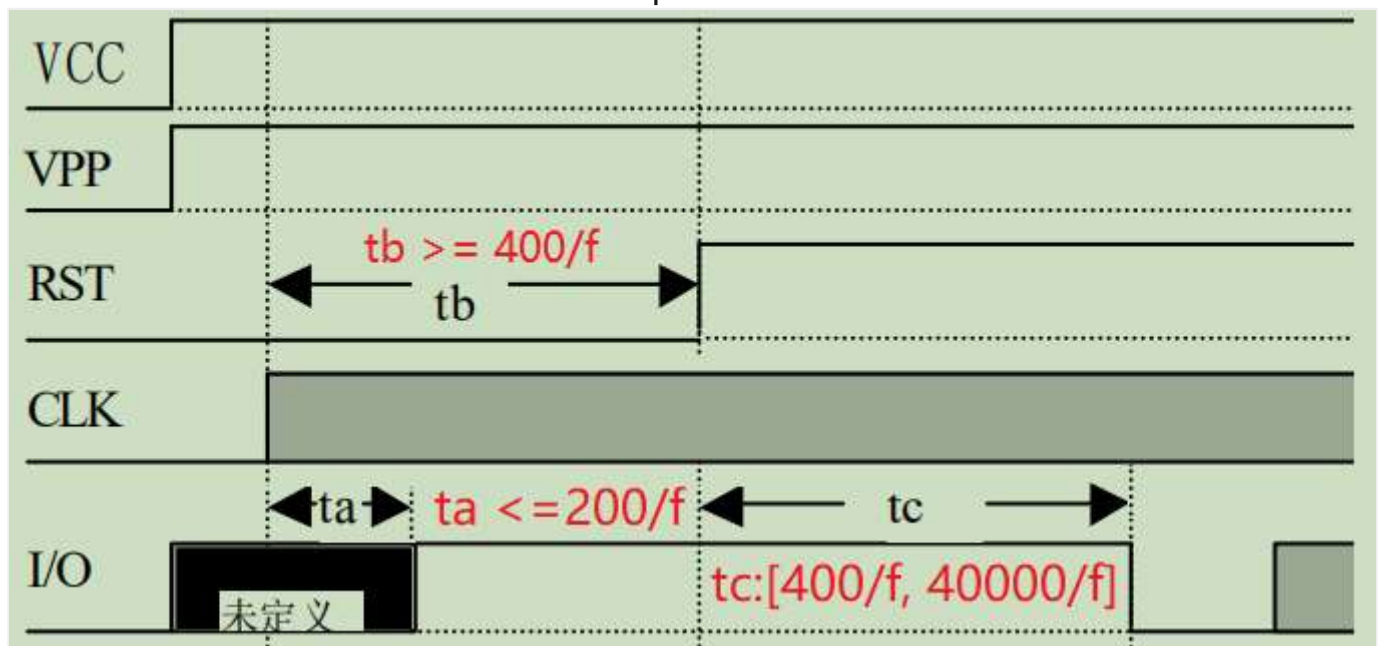
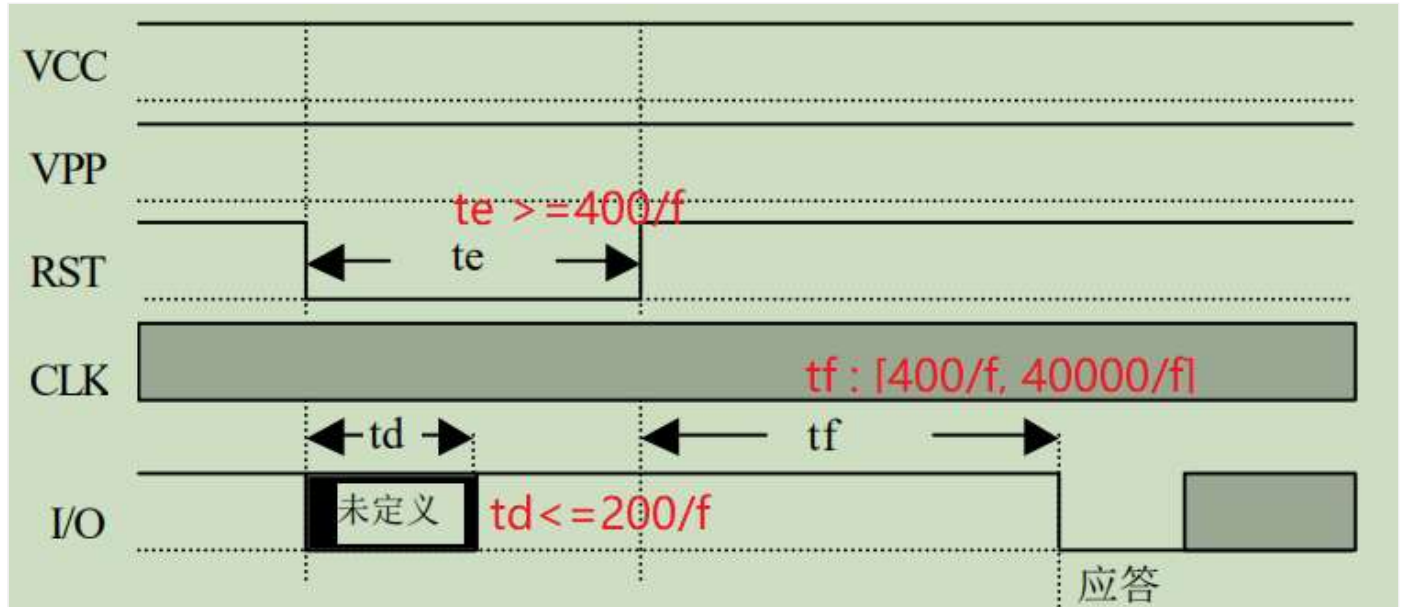


Figure 4 Activation and cold reset

4.3 Warm reset

- 1) When VCC remains powered and CLK provides a suitable and stable clock signal.
- 2) RST is placed in state L for at least 400 clock cycles to initiate a warm reset.
- 3) The IO interface sets IO to state H within 200 clock cycles after RST state L.
- 4) Within 400 to 40,000 clock cycles after RST sets the H state, I/O will receive a reset response.
- 5) If more than 40,000 clock cycles after the RST sets the H state, and the I/O reset

response does not start, the warm reset fails and the interface releases the circuit.



The difference between cold reset and warm reset is, Cold reset is initiated after power-on again and a stable clock signal is added, while warm reset is initiated by the RST signal when the power and clock signals are maintained.

4.4 Clock stop

- 1) I/O remains in state Z for at least 1860 clock cycles (t_g);
- 2) When the clock is stopped (from T_e to T_f), CLK should remain in state H or state L, which is specified by parameter X.
- 3) After T_f , the interface device restarts the clock, and the information exchange on the I/O must be at least 700 clock cycles later (t_h).

4.5 Release

- 1) RST is set to state L
- 2) CLK is set to state L
- 3) VPP release (if activated)
- 4) I/O is set to state A
- 5) VCC release

5 ATR

ATR is answer to reset (answer to reset). After the reader resets the card, the card returns the first instruction of the reader. The reset reply format is as follows:

TS T0 TA1,TB1,TC1,TD1,...,TAi,TBi,TCi,TDi T1,T2,...,TK TCK

5.1 Initial character TS

- TS: Initial character, which defines the encoding protocol of data bytes in all subsequent characters, followed by up to 32 characters.
- 3B: Forward agreement; 3F: Reverse agreement.

5.2 Format character T0

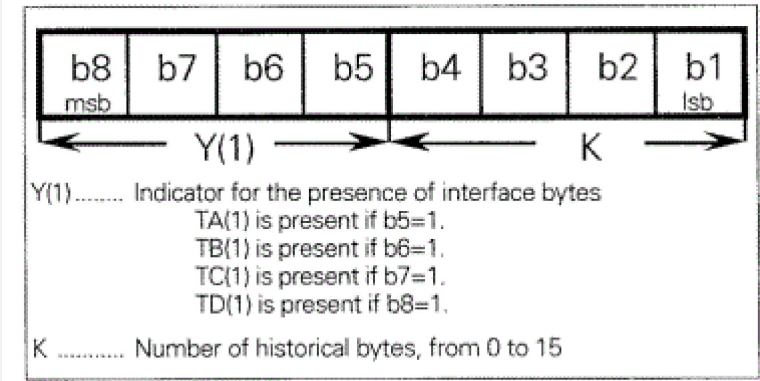


图 3 T0 编码

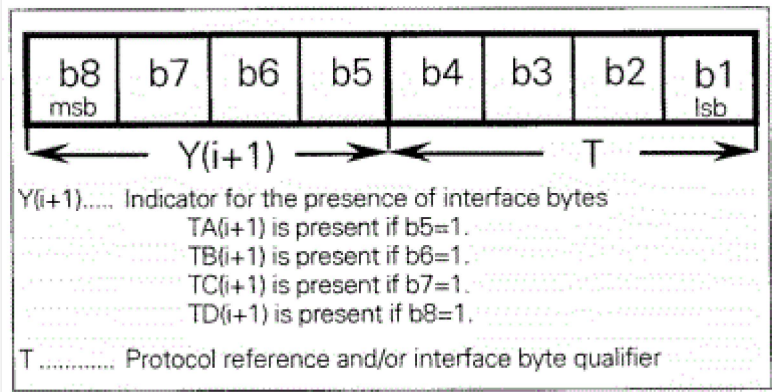


图 4 TD(i)编码

- Y1 (m7~m4): mark whether TA1, TB1, TC1, and TD1 exist;
- K (m3~m0): the data length of historical characters, usually T=0 (asynchronous half-duplex character transmission protocol), T=1 (asynchronous half-duplex block transmission protocol).
- **Basic response:** If only T=0 is used, the IC card will return T0="6X", indicating the existence of characters TB1 and TC1. If only T=1 is used, the IC card returns T0="EX", indicating the existence of characters TB1 to TD1. The value of "X" indicates the number of historical characters transmitted.
- **Terminal reaction:** If the T0 returned by the IC card is any value, as long as it is correctly specified and consistent with the interface characters TA1 to TD1 and the actual returned historical byte, the terminal shall not refuse.

Table 1 Characters sent back in response to reset with T=0

character	value	Remarks
TS	"3B" or "3F"	Indicates a forward or reverse agreement
T0	"6X"	TB1 and TC1 appear, X indicates the number of historical bytes
TB1	"00"	Not applicable to VPP
TC1	"00" to "FF"	Indicates that additional protection time is required, and the data "FF" has a special meaning

Table 2 Characters sent back in response to reset with T=1

character	value	Remarks
TS	"3B" or "3F"	Indicates a forward or reverse agreement
T0	"EX"	TB1 and TD1 appear, X represents the number of historical bytes

character	value	Remarks
TB1	"00"	Not applicable to VPP
TC1	"00" to "FF"	Indicates that additional protection time is required, and the data "FF" has a special meaning
TD1	"81"	Use T=1 protocol, TA2 to TC2 do not exist, TD2 exists
TD2	"31"	Use T=1 protocol, TA3 to TB3 exist, TC3 and TD3 do not exist
TA3	"10" to "FE"	Return IFSI, indicating the initial value of the IC card information field size, 16~254 bytes
TB3	High 4-bit group "0" to "4"; Low 4-bit group "0" to "5"	BWI = 0~4; CWI = 0~5
TCK		Check character

Note: Comparison of T=0 and T=1:

T=0 is the smallest transmission unit with characters. In this protocol, the card reader sends an APDU command to the card, which is composed of 5 characters, each of which has a clear and fixed meaning. In the T=1 protocol, in addition to defining the structure of the block, other information is stored in the INF, which means that the instructions and data sent by the card reader to the card are stored in this INF domain, and there is no response to these instructions and The data is strictly regulated. In this regard, T=1 is more flexible than T=0.

5.3 Interface characters T_{Ai}, T_{Bi}, T_{Ci}, T_{Di} (i=1, 2, 3, etc.)

T_{Ai}: High 4bit is FI (clock rate conversion factor), low 4bit is DI (baud rate correction factor);

T_{Bi}: bit₁~bit₅: Maximum programming voltage, bit₆~bit₇: Maximum programming current, bit₈ is not used.

T_{Ci}: Calculate the reference of eight extra guard time.

T_{A2}: If it exists, it is the dedicated mode, and if it does not exist, it is the negotiation mode. The protocol needs to be confirmed through the PPS command.

F_i and D_i are used when bit₅=0 of T_{A2}, and the default value is used when bit₅=1.

T_{Di}: Indicate the protocol type and whether there are subsequent interface characters. T_{Di} includes two parts, Y_{i+1} and T. Y_{i+1} is the upper four digits, which respectively represent the subsequent interface characters T_{Ai+1}, T_{Bi+1}, T_{Ci+1}, T_{Di+1}. If 1 exists, T is the lower four digits, indicating the type of protocol to be sent subsequently.

接口字符	编码条件	b7~b0	说明
TA1	B	X X X X X X X X	可回送，缺省值为F=372，D=1
TB1	B	0 0 0 0 0 0 0 0	VPP未连至IC卡
TC1	B	X X X X X X X X	额外保护时间N值
TD1	T=1	1 0 0 0 0 0 0 1	T=0时，可不回送TD1
TA2	专用接口字符	-----	协商方式（非专用）时，可不回送TA2
TB2	----	-----	不得回送，因为是表示VPP值
TC2	T=0	-----	传达WI用，IC卡不得回送，缺省值10
TD2	T=1	0 0 1 1 0 0 0 1	T=0时，不得回送
TA3	T=1	X X X X X X X X	"00"~"F"，及"FF"禁止使用
TB3	T=1	0 X X X 0 Y Y Y	XXX在000~100，YYY在000~101内
TC3	T=1		可不回送，缺省表示采用LRG校验

5.3.1 TA1

TA1 conveys the values of FI and DI, among them: FI (higher 4 bits) is used to determine the value of F, F is the clock frequency conversion factor, used to modify the clock frequency provided by the terminal after the answer to reset. DI (lower 4 bits) is used to determine the value of D, called the bit rate adjustment factor, used to adjust the bit duration used after the response to reset.

表 7 — Fi, 指明的时钟率转换因子的值

FI	0000	0001	0010	0011	0100	0101	0110	0111
Fi	372	372	558	744	1116	1488	1860	RFU
f (max) MHz	4	5	6	8	12	16	20	—

RFU=留作未来使用

FI	1000	0001	0010	0011	0100	0101	0110	0111
Fi	RFU	512	768	1024	1536	2048	RFU	RFU
f (max) MHz	—	5	7,5	10	15	20	—	—

表 8 — 指明的波特率校正参数的值

DI	0000	0001	0010	0011	0100	0101	0110	0111
Di	RFU	1	2	4	8	16	32	RFU

DI	1000	1001	1010	1011	1100	1101	1110	1111
Di	12	20	RFU	RFU	RFU	RFU	RFU	RFU

• The bit duration at the time of answer to reset is called the initial et_u , and is formally calculated:

The initial $et_u = 372/f$ second, where the unit of f is Hz, which represents the initial frequency of the reset response.

• The duration after the answer to reset (and the establishment of global parameters F and D) is called the current et_u , and its calculation formula is:

The current $et_u = F/Df$ seconds, where the unit of f is Hz, which represents the current operating frequency during subsequent transmission.

The default values used during the response to reset: FI=1 and DI=1, meaning: F=372 and D=1

• **Basic response:** If the IC card does not return TA1, the default values F=372 and D=1 will continue to be used throughout the subsequent information exchange process.

• **Terminal reaction:** The terminal shall not reject an IC card with TA1="11" (if the b5 bit of T0 is set to 1), and continue to use F=372, D=1 in all subsequent transactions.

5.3.2 TB1

TB1 transmits the values of PI1 and I1, where:

– Defined in bits b1~b5, used to determine the maximum programming voltage P value required by the IC card. b1=0 means that the IC card does not use V_{pp} .

– Defined in bits b6~b7, used to determine the maximum programming current I value required by the IC card. b1=0 means that this parameter is not used.

– The b8 bit is not used and set to logic "0".

• **Basic response:** The IC card will return TB1 = "00", which means that the IC card does not use VPP.

• **Terminal response:** If b6 of T0 is set to "1" and the TB1 returned by the IC card is any value, or b6 of T0 is set to "0", when the IC card returns TB1, the terminal shall not

reject the card, but it does not generate VPP, and Continue the process of using the card, just like sending back TB1="00".

5.3.3 TC1

TC1 transmits the value of N, and N is the extra guard time. N is b8 of TC1_{b1-bit binary code}, its value represents the number of etu added by the extra guard time, and its value is 0 Between 255. N=255 (TC1="FF") has a special meaning, it indicates the shortest time interval between the leading edges of the start bits of two adjacent characters: T=0 is 12 etu; T=1 is 11 etu.

Note that TC1 is only used for the timing between two adjacent characters sent from the terminal to the IC card. It is not used for the timing between two adjacent characters sent from the IC card to the terminal, nor is it used for the reverse transmission between two characters. (Please refer to the description of the timing of T=0 or T=1 in the section Transmission Protocol).

- **Basic response:** The value of TC1 returned by the IC card should be in the range of "00" to "FF".

- **Terminal response:** When the IC card does not send back TC1 (assuming that the b7 bit of T0 is set to "0"), the terminal shall not refuse, and

The processing of the card renewal process is like sending back TC1="00".

It is recommended that when designing an IC card, TC1 should be set to the minimum acceptable value of the IC card. A larger TC1 value will cause the communication between the terminal and the IC card to be too slow, thereby prolonging the transaction time.

5.3.4 TD1

TD1 indicates whether there is more interface byte transmission and information about the type of transmission protocol.

- The upper nibble is used to indicate whether the characters TA2 to TD2 exist. If the bits b5 to b8 are set to logic "1", it indicates the existence of the character in the corresponding TA2 to TD2.

- The lower nibble provides information about the type of transport protocol used by subsequent exchanges.

Basic response: When the T=0 protocol is selected, the IC card will not send back TD1, and the subsequent transmission protocol defaults to T=0. When the T=1 protocol is selected, the IC card must send back TD1="81" to indicate the existence of TD2, and the subsequent transmission protocol is T=1.

Terminal reaction: When the high quartet of TD1 returned by the IC card has any value (assuming that the returned value is correctly represented and consistent with the actual returned interface characters TA2 to TD2), and the value of the low quartet is "0" or "1", The terminal must not refuse. The terminal must refuse to send back other IC cards with the value of TD1.

5.3.5 TA2

The presence or absence of the character TA2 correspondingly indicates that the IC card is working in a specific mode or an interactive mode.

Basic response: The IC card UQF does not send back TA2. The absence of TA2 means it works in interactive mode.

Terminal response: If the terminal can support the additional conditions specified by the IC card through TA2 during the response to reset, it will not reject such IC cards and should be able to use these conditions immediately.

5.3.6 TB2

TB2 transmits PI2, and PI2 is used to determine the value of the programming voltage P required by the IC card. When it exists, it replaces the value of PI1 returned by TB1.

Basic response: IC card should not return TB2.

Terminal reaction: The terminal should not refuse the IC card to send back TB2, but the terminal will not generate VPP regardless of whether it is sent back or not and what value is sent back.

5.3.7 TC2

TC2 is unique to the T=0 protocol. It conveys the difference between the rising edge of the start bit of any character sent by the IC card and the rising edge of the start bit of the previous character sent by the IC card or terminal. The maximum working waiting time (WI). The working waiting time is: $960 \times D \times WI$.

Basic response: The IC card must not send back TC2, and the default value WI=10 is used in subsequent communications.

Terminal reaction: The terminal shall not refuse to send back IC card with TC2=10.

5.3.8 TD2

TD2 indicates whether to send more interface bytes, and the protocol type used for subsequent transmissions.

- The upper nibble is used to indicate whether the characters TA3 to TD3 exist, and the bits b5 to b8 are set to logic "1", which means the corresponding TA3 to TD3 exist.
- The lower nibble indicates the type of protocol used for subsequent transmission. If T=1 is used, the value of the lower nibble is "1".

Basic response: If T=0 is used, the IC card does not send back TD2, and the subsequent transmission protocol defaults to T=0. If T=1 is used, and the IC card must return TD2="31" to indicate the existence of TA3 and TB3, and the subsequent transmission protocol is T=1.

Terminal reaction: When the upper nibble of TD2 returned by the IC card has any value (assuming the returned value is correct and consistent with the actual return interface characters TA3 to TD3), and the lower nibble is "1" or "E", then The terminal must not refuse, and the terminal must refuse to return IC cards with other TD2 values.

5.3.9 TA3

TA3 returns the information field length integer (IFSI), which determines the IC card information field length (IFSC) and specifies the maximum length of the word group that the IC card can receive. It expresses the length of IFSC in byte form, and the value range is any value between "01" and "FE". The two values "00" and "FF" are reserved for spare.

Basic response: If T=1 is used, the IC card must return TA3 as a value between "10" and "FE" to indicate that the initial IFSC is within the range of 16 to 254 bytes.

Terminal reaction: The terminal must not reject an IC card that has not sent back TA3 (assuming that the b5 bit of TD2 is set to "0"), but if it accepts such an IC card, it must continue the card use process with "20" as the value of TA3. The terminal must reject the returned IC card whose TA3 value is between "00" and "0F" or is "FF".

5.3.10 TB3

TB3 indicates the value of CWI and BWI used to calculate CWT and BWT. TB3 consists of two parts, the lower nibble ($b1_{b4}$) is used to represent the value of CWI, and the high nibble ($b5_{b8}$) is used to indicate the value of BWI.

Basic response: If T=1 is used, the IC card should send back TB3 like this: The high nibble is '0' '5', the lower nibble takes the value '0' '4'. That is, the value of CWI is between 0 and 5, and the value of BWI is between 0 and 4.

5.3.11 TC3

TC3 specifies the type of block error detection code used, the type is represented by the b1 bit, and b2 to b8 are not used.

Basic response: When using Longitudinal Redundancy Check (LRC) as an error detection code, the IC card does not need to send back TC3.

5.4 Historical characters T1, T2, Tk

Indicate some general information, such as the manufacturer of the card, the chip embedded in the card, the file status of the card, etc.

5.5 Check Character Tck

TCK has a value to check the integrity of the data sent during the response to reset. The value of TCK should make the XOR operation result of all bytes from T0 to TCK (including TCK) to 0.

Basic response: If T=0 protocol is used, TCK will not be sent, and in other cases, TCK will be sent.

Terminal reaction: When using the T=0 protocol, the terminal should refuse to send back the TCK IC card. If the IC card returns TCK, the terminal should be able to assign a value to TCK.

6 ISO7816 communication data example

In this example, Kingst LA5016 usb logic analyzer of Kingst Electronics is used to detect IOS7816 data communication. The following pictures are screenshots of a complete packet analysis. It can be seen from the figure that the IOS7816 protocol can parse ATR into initial characters TS, format characters T0/T1, interface characters T_{Ai}~T_{Di}, historical characters T1, T2,..., Tk, and check characters Tck, etc.

Protocol analysis parameter settings:

